

Roudaina Kouka

Software Engineering Student | Cloud-Native, DevOps & AI

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PROFILE

Final-year Software Engineering student specializing in **Cloud**, **DevOps**, and **DevSecOps**, with hands-on experience building **CI/CD pipelines**, **cloud infrastructure**, and automation workflows. Conducting NLP/ML **research** while developing **agentic-AI** and cloud-native software solutions using AWS, Azure, GCP, Docker, and Kubernetes. **Co-founder & CMO** of **CarthaPlay**.

PROFESSIONAL EXPERIENCE

DevOps & Software Engineering Intern, [Inetum Tunisia](#), [Microsoft Department](#) 07/2026 – 08/2026 | Tunisia

- **Designed** and **deployed** three Azure **DevOps** Server **CI/CD** pipelines for Dynamics 365/Dataverse, data migration, and .NET 8 API deployment to IIS, reducing deployment time by **40%** and enabling fully offline, on-premises releases.
- **Automated** validation, packaging, backup, restoration, health checks, and **rollback** across all **pipelines**.
- **Built** an internal **.NET 8** developer tool (CRM ALM dashboard) centralizing deployment management, **Git-based versioning**, solution analysis, and pipeline orchestration.
- **Tools**: Azure DevOps Server, CI/CD, YAML, PowerShell, .NET 8, C#, ASP.NET Core, IIS, Dynamics 365, Git, SonarQube, REST APIs

Software Engineering Intern — Cloud & DevOps, [Pulse](#), [Software-OS Department](#) 06/2026 – 09/2026 | U.S. (Remote)

- **Provisioned** a **cloud-native** data platform on **GCP** with Terraform, including **BigQuery**, Cloud Storage, IAM, Secret Manager, Artifact Registry, Cloud Run, and Cloud Scheduler.
- **Built** a secure data platform with **CI/CD**, GCS lifecycle policies, multi-source ingestion, BigQuery partitioning, Redis caching, least-privilege IAM, and automated disaster recovery.
- **Tools**: GCP, Terraform, BigQuery, Cloud Run, IAM, Secret Manager, Redis, CI/CD, IaC

Full Stack Developer Intern, [Pixemantic](#), [Software Department](#) 06/2025 – 09/2025 | Tunisia

- **Implemented** AI-assisted medical note generation, a chatbot assistant, and OCR-based document processing for App4Doctors, a MERN-stack telemedicine platform.
- **Tools**: React, Node.js, Express, MongoDB, OCR, REST APIs

Marketing Intern, [Abinway](#), [Marketing Department](#) 10/2024 – 03/2025 | Germany (Remote)

- **Contributed** to the North African Marketing Team by supporting outreach and communication with prospective candidates interested in **international** mobility opportunities.
- **Developed** targeted content for students and young professionals while thriving in a multicultural remote environment, strengthening **communication**, adaptability, and **teamwork**.

RESEARCH EXPERIENCE

Research Intern – Cybersecurity NLP, *Faculty of Sciences of Tunis, University of Tunis El Manar* 05/2026 – Present | Tunisia

- **Design** an NLP pipeline for **event extraction**, **temporal ordering**, and **causal relation classification** in CVE vulnerability descriptions.
- **Constructed** and manually validated an annotated dataset using **Label Studio**, covering 383 CVE descriptions.
- **Trained** and **evaluated** a **BERT-based binary classifier** for causal event relations after comparing alternative relation-classification formulations.
- **Co-author** a research manuscript targeting a peer-reviewed **AI/Security/NLP** venue.
- **Tools**: Python, NLP, Machine Learning, CVE/NVD, Label Studio

Research Contributor – Reinforcement Learning for Adaptive Learning, *CarthaPlay* 01/2024 – 04/2026 | Tunisia

- **Explored** a **reinforcement-learning** approach that uses students' answers and in-game actions to assess learning progress and adapt educational content.
- **Designed** an adaptive questioning mechanism that selects and reformulates questions based on students' responses and estimated learning state.
- **Shaped** AI-driven **educational games** combining student assessment, personalized feedback, and adaptive learning.
- **Tools**: Python, Reinforcement Learning, Machine Learning, React.js, JavaScript

EDUCATION

Bachelor's Degree in Software Engineering (BSE), *Faculty of Sciences of Tunis* 09/2022 – Present | Tunis, Tunisia

- **Passed and ranked 10/60**
- **Relevant Courses**: Cloud Computing, Distributed Systems, Computer Systems Design, Machine Learning & AI, Cybersecurity, Big Data, DevOps & Microservices

Integrated Pre-Engineering Program, *Faculty of Science of Tunis* 09/2022 – 05/2024 | Tunis, Tunisia

- **Passed and ranked 14/120**
- **Relevant courses**: Mathematics, Databases, Operating Systems

Baccalaureate Degree, *Abou Kacem Chebi, Kesra* 07/2022 | Siliana, Tunisia

- **Top regional ranking: 1/283**

SKILLS

Software Engineering & Programming: Python, Java, C#, JavaScript, Node.js, C++, Go, Bash
Cloud & Cloud-Native: AWS, Azure, GCP, Docker, Kubernetes, Microservices, Cloud-Native Architecture, REST APIs, Linux
AI & Machine Learning: Machine Learning, NLP, BERT, Reinforcement Learning, LLM APIs, LLMs, RAG, AI Agents, MLOps/LLMOps
DevOps & Automation: CI/CD, Azure DevOps, Jenkins, GitHub Actions, Terraform, Ansible, ArgoCD, PowerShell, IaC
DevSecOps & Security: DevSecOps, Software Supply-Chain Security, OPA/Rego, TLS/mTLS, Security Auditing, SonarQube, Trivy
Soft Skills: Analytical Thinking, Problem-Solving, Team Collaboration, Communication, Adaptability, Technical Report Writing, Research Literature Review
Monitoring & Tools: Prometheus, Grafana, Git, GitHub, Jira, GitLab CI/CD

PROJECTS

Cloud-Native DevOps & CI/CD Platform Deployment

- **Designed and deployed** a scalable three-tier web application using ReactJS, NodeJS, and MongoDB on **AWS EKS**, implementing a cloud-native architecture with containerized services.
- **Automated infrastructure provisioning and CI/CD** using **Terraform, Jenkins, Docker, and Amazon ECR**, enabling repeatable infrastructure deployment and automated application delivery.
- **Implemented GitOps-based deployment and observability** using **ArgoCD, Kubernetes, Prometheus, and Grafana**, supporting automated synchronization, monitoring, and operational visibility.
- **Key Technologies:** AWS, Jenkins, Docker, Kubernetes, SonarQube, Trivy, ArgoCD, Prometheus, Grafana, ReactJS, NodeJS.

Agentic AI Platform for Intelligent Insurance Engagement

- **Architected** an **Agentic AI** platform for BH Assurance, orchestrating conversational workflows, contextual reasoning, intent analysis, product recommendations, and automated client engagement across multiple channels.
- **Implemented** an AI agent architecture integrating **LLMs, RAG, hybrid intent analysis, and modular FastAPI services**, enabling autonomous, context-aware interactions grounded in insurance and legal knowledge.
- **Built** a scalable event-driven ecosystem integrating **Kafka, Gmail API, MongoDB, and React/TypeScript**, supporting automated email processing, real-time communication, and AI-driven workflows.
- **Key Technologies:** Python, LLMs, **Agentic AI**, RAG, FastAPI, React, TypeScript, Node.js, MongoDB, Kafka, Docker.

Netflix Clone Deployment with DevSecOps

- **Automated** end-to-end **DevSecOps** delivery for a containerized **Netflix** clone on **AWS** using **Jenkins** and **Docker**.
- **Integrated automated security and quality controls** using **SonarQube, OWASP Dependency-Check, and Trivy**, performing source-code, dependency, filesystem, and container-image vulnerability analysis within the CI/CD workflow.
- **Deployed and monitored** the application on **Kubernetes**, using **ArgoCD for GitOps-based delivery** and **Prometheus/Grafana for real-time observability**.
- **Key Technologies:** ReactJS, Docker, Kubernetes, AWS EC2/EKS, Jenkins, SonarQube, Trivy, Prometheus, Grafana, Helm, DockerHub

AI-Based Automatic Plant Leaf Disease Detection & Classification

- **Created** an AI-based plant disease detection system combining **Computer Vision, Machine Learning, and Deep Learning** for automated image-based diagnosis.
- **Designed** an end-to-end image processing pipeline covering preprocessing, segmentation, feature extraction, and hybrid ML/DL classification, integrating **intelligent decision-making agents** for automated diagnosis.
- **Built** a full-stack interface enabling image upload and real-time visualization of model predictions.
- **Key Technologies:** Python, Deep Learning, Machine Learning, OpenCV, ReactJS, NodeJS, ExpressJS, MongoDB.

CERTIFICATIONS

AWS Certified Solutions Architect — Associate (SAA-C03)
AWS - Cloud Architecting — Training Badge
AWS Certified Cloud Practitioner
Certified Cybersecurity Educator Professional (CCEP)
Certified Red Team Operations Management (CRTOM)
CompTIA Security+
AI and Machine Learning — Virtual University of Tunis
Python Data Associate Certification — DataCamp

ACADEMIC & LEADERSHIP ACTIVITIES

IEEE CS Chapter FST, DevOps Instructor	10/2025 – Present
Optima Junior Enterprise, Senior Business Development, R&D	10/2024 – Present
ATIA Club FST, Vice President	09/2025 – Present

AWARDS & DISTINCTIONS

1st Place - AI Night challenge , Association of Scientific Research & Innovation in Computer Science (ARSII)	06/2026
1st place - Next BH hackathon, BH Insurance	04/2026
1st Place – D-Clickathon, Digital Learning Lab (DAAL)	01/2026

LANGUAGES

Arabic — Native | French — Bilingual | English — B2 | German — B1